Mukul Prasad

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Education

Ramaiah Institute of Technology, Bengaluru, India — Expected Jun 2028
B.E. in Computer Science & Engineering (Artificial Intelligence & Machine Learning) — CGPA: 9.47

Experience

ML Intern | *FlyRank AI* — July 2026 – Present
- Selected as an ML Intern to work on machine learning research, model development, experimentation, and AI-driven product features using Python, PyTorch, Scikit-learn, and modern LLM workflows.

Open Source Mentor | *GirlScript Summer of Code (GSSoC'26)* — Jun 2026 – Present
Selected among 135 mentors nationally
- Reviewed 20+ pull requests with structured feedback on code quality, logic, and documentation standards.
- Guided 15+ contributors in Python, JavaScript, and REST APIs to successfully merge first open-source contributions.

Research Intern, AI/ML Drug Repurposing | *IEEE Computer Society* — Feb 2026 – Present
- Building an AI-driven drug repurposing pipeline using PyTorch Geometric, RDKit, and biomedical knowledge graphs spanning **14K+ drugs**, **20K+ diseases**, and **100K+ interactions**.
- Co-authoring an **IEEE research paper** by developing the **4-component NDPI** framework integrating GNN predictions, PubMed literature mining, Morgan fingerprints, and WHO equity metrics.

Projects

DonorConnect — Intelligent Blood Donor Matching — *Top 10/1,000 Teams — ECE Society BIT Mesra Hack*
Python, TensorFlow, TensorFlow Lite, Firebase, REST APIs — Github | Demo | Live
- Trained a deep learning donor-matching model on **50K+ donor-recipient records- 94% compatibility accuracy**.
- Optimized inference to **<300 ms** using TensorFlow Lite and deployed on Firebase .

KisaanMitra — Multi-Agent Agricultural AI — *1st Runner-up — Agentic AI Hackathon*
Python, Multi-Agent Systems, NLP, AWS Lambda, Deep Learning — Github | Demo
- Engineered a **5-agent** MAS for crop advisory, market intelligence, weather monitoring, IoT integration, and alerting.
- Trained on **12K+ agricultural records** achieving **87% accuracy**; deployed serverlessly via AWS Lambda with **sub-2 s** inference.

Project Astraeus — AI Mission Control — *SIH 2025 Merit*
Python, TensorFlow, AWS Lambda, Digital Twin Architecture — Github | Demo | Live
- Built a real-time digital twin mission-control platform with **91% predictive failure detection accuracy**.
- Designed AWS Lambda REST pipelines reducing manual operational intervention by **35%**.

RegGraph-AI — Regulatory Graph Intelligence — *Zonal Champion, Finalist — HackArena 2.0, IIIT Delhi*
Python, Graph AI, NLP, Deep Learning, RAG — Github | Demo
- Built a Graph AI pipeline for regulatory intelligence using knowledge graphs, NLP entity extraction, RAG. Delivered explainable compliance retrieval by surfacing regulatory risk pathways from large document corpora.

Technical skills

Languages: Python, C,C++, Java, SQL
AI / ML: PyTorch, TensorFlow, Scikit-learn, Hugging Face, LangChain, LangGraph, NLP, RAG Pipelines, Multi-Agent Systems, Deep Learning
Web & Backend: Node.js, MERN Stack, REST APIs, Firebase
Tools: Git, GitHub(1.5k+ contributions), MCP(Integration and Operations)
Cloud & MLOps: AWS (Lambda, S3, SageMaker, EC2, IAM), Docker, CI/CD, Model Deployment
Databases: PostgreSQL, MongoDB, MySQL, Redis, Supabase, Hive, Hbase, Graph DB(Neo4j)
Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems,Complier Design

Achievements

- **ML Competitions: Top 150/10,760** (as of Jul 2026) - Kaggle Wellbore Geology Prediction(International team led by PyJa) | Top 25/348 - Kaggle Formula 1 Race Result Classification (Polaris Institute of Technology)
- **Hackathons:** 1st runner-up - GenAI Fusion'25 Hack | Zonal Champion - HackArena 2.0 (IIIT Delhi) | Top 10/1,000 - BIT MESRA Hack'24 | Finalist - Samsung PRISM'25,'26, wHACKiest'26, Hack-A-War'25 (Top10), Codecrafter'26, HackMalenadu'26 | SIH'25 Merit
- **Academic:** Nearly perfect **9.95 SGPA** (Sem II) | 9.61 SGPA (Sem III) - Ramaiah Institute of Technology
- **Certifications:** AWS Cloud Practitioner Essentials | AWS Fundamentals of AI/ML | IBM India Impact GenAI Program - IISc Bengaluru
- **Leadership & POR:** Leading MSRIT'28 & MSRIT'29 communities (**3.5K+ active members**) | Founding Core Member - Amazon Web Services (AWS) Cloud Club | Core Member - Quantum Cluster (QpiAI Industry MoU) | Webmaster - IEEE Computational Intelligence Society | DeepStation (US-based) | E-Cell & DSC (formerly GDG)